Team Preparation Guide

NLP Midsem Evaluation — CS F429
Track A: Dynamic Uncertainty-Aware Attribution

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This guide covers everything you need to know for tomorrow.
Read it fully before the evaluation session.

1. What Is This Project — The Simple Version

The Problem

AI language models like GPT sometimes make up information that is not supported by their sources. This is called hallucination. For example, if you ask "When did Anne Frank die?" and the AI has context saying "February 1945" but answers "February 7, 2022" — that is a hallucination.

> **Our project builds a system that automatically detects when an AI is hallucinating.**

Our Solution — In One Sentence

We measure how much a language model's uncertainty changes when retrieved context is added. If the uncertainty does not drop, the model is likely hallucinating.

How It Works — Step by Step

Step	What Happens	Why
1	Take a question, retrieved context, and AI answer	These are the inputs to our pipeline
2	Run GPT-2 on the answer WITHOUT context	Get baseline uncertainty at each token
3	Run GPT-2 on the answer WITH context	Get uncertainty when context is provided
4	Measure the difference between both runs	This difference is our hallucination signal
5	Combine into a composite score	Single number: high = hallucinated, low = faithful

The Key Insight

> *If you give someone a fact they already know, their confidence increases. If they made something up, giving them the real fact does not change their confidence because they are not using it. We apply this same logic to the AI model.*

What We Used

Component	What It Is	Why We Used It
GPT-2	Small language model (124M params)	Gives us token-level probability distributions. Runs on CPU.
RAGTruth	Primary dataset (15,090 samples)	Has word-level hallucination annotations from 6 LLMs
HaluEval	Secondary dataset (10,000 samples)	Used for cross-domain transfer testing (Experiment 4)
Logistic Regression	Composite model	Combines 4 metrics into one hallucination score
HuggingFace	Model and dataset library	Used to load GPT-2 and both datasets

2. The Five Metrics — What They Are and What They Mean

Metric 1: Entropy Only (Baseline 1)

What it is: Shannon Entropy measures how uncertain the model is about the next token without any context.

```
Formula: H(p) = -SUM [ p(x) * log(p(x)) ] for all tokens x
```

Simple explanation: If entropy is high, the model is confused. If entropy is low, the model is confident. We use this as our first baseline.

Our result: AUROC = 0.5806 (barely better than random guessing)

Metric 2: Information Gain (IG)

What it is: How much entropy DROPS when context is added.

```
Formula: IG(t) = H(p_no_context)(t) - H(p_with_context)(t)
```

Simple explanation: Positive IG = context helped the model = faithful. Negative IG = context confused the model = possible hallucination.

Our result: AUROC = 0.2963 individually (but contributes to composite)

Metric 3: KL Divergence — Our Strongest Metric

What it is: How DIFFERENT the entire probability distribution is between the no-context and with-context runs.

```
Formula: KL(P||Q) = SUM [ P(x) * log(P(x)/Q(x)) ]
```

Simple explanation: KL measures the full shape change of the probability distribution. Unlike entropy which only measures spread, KL captures exactly how each token probability shifted.

Our result: AUROC = 0.7235 — our best individual metric

THE EVALUATOR WILL ASK YOU TO EXPLAIN THIS AND SHOW THE CODE.

The code for KL divergence is in notebook 02_core_pipeline inside the compute_all_metrics function:

```
for p_no, p_with in zip(probs_no_ctx, probs_with_ctx): p_no = p_no + 1e-10 p_no = p_no /
p_no.sum() # normalize p_with = p_with + 1e-10 p_with = p_with / p_with.sum() #
normalize kl = float(np.sum(kl_div(p_no, p_with))) # KL for this token
kl_per_token.append(kl) kl_divergence = float(np.mean(kl_per_token)) # average across
tokens
```

Metric 4: Confidence Drop

What it is: How the model's maximum token probability changes when context is added.

```
Formula: ConfDrop(t) = max(P_no_context)(t) - max(P_with_context)(t)
```

Simple explanation: If the model becomes less confident when given context, it may mean the context contradicted what the model wanted to say — a hallucination signal.

Our result: AUROC = 0.2859 individually

Metric 5: Semantic Entropy

What it is: Entropy computed over only the top-10 most probable tokens instead of all 50,257.

Simple explanation: Captures meaning-level uncertainty — is the model confused between semantically similar options?

Our result: AUROC = 0.2734 individually

The Composite Score

We combine all 4 metrics (we dropped semantic entropy from the composite as it hurt performance) using Logistic Regression trained on 300 training samples:

```
Features used: entropy_only, information_gain, kl_divergence, confidence_drop
```

Result: Composite AUROC = 0.7345 — beats both baselines and closes 53.2% of the SOTA gap.

3. How to Present Each Experiment — What to Say Out Loud

Experiments 1 & 2 — Main AUROC Table

What to say:

> *"We ran all five metrics individually on the RAGTruth test set — 500 samples, 250 hallucinated and 250 clean, drawn from the held-out test split. We then built the composite incrementally by adding each metric. The full composite achieves an AUROC of 0.7345 with a 95% bootstrap confidence interval of 0.689 to 0.779. This beats both our baselines — entropy only at 0.58 and SelfCheckGPT at 0.68. KL divergence is our strongest individual metric at 0.72."*

Key numbers to remember:

Composite AUROC: 0.7345 | CI: [0.689-0.779] | Beats entropy (0.58) and SelfCheckGPT (0.68)

Experiment 3 — Temporal Precedence

What to say:

> *"Experiment 3 checks whether our metrics show a signal before the hallucination happens. We used the actual character-level span annotations from RAGTruth to find the exact token position of each hallucination — we call this position t. We then measured our metrics at t minus 3, t minus 2, t minus 1, t, and t plus 1. We found that KL divergence shows an elevated signal at t minus 3, which is 3 tokens before the hallucination — suggesting our metric can detect hallucination before it occurs. The Mann-Whitney U test did not reach significance, which is a limitation we acknowledge."*

Key numbers to remember:

KL at t-3: 10.7348 | KL at t: 10.3816 | n=141 valid samples | p=0.1117 (not significant)

Experiment 4 — Cross-Domain Transfer to HaluEval

What to say:

> *"Experiment 4 tests whether our pipeline generalizes to a completely different dataset. We ran our full composite on HaluEval QA samples with zero retraining or recalibration — pure zero-shot transfer. Our composite actually improved on HaluEval, achieving 0.9249 AUROC compared to 0.7345 on RAGTruth. This suggests our uncertainty-based approach generalizes well across domains. The rank ordering of metrics remained stable across both datasets."*

Key numbers to remember:

HaluEval AUROC: 0.9249 | RAGTruth AUROC: 0.7345 | Drop: -0.1904 (improved) | n=200 samples

Experiment 5 — Hallucination Type Breakdown

What to say:

> *"RAGTruth labels each hallucination as either contradictory — where the AI directly conflicts with the context — or unsupported — where the AI introduces information not in the context. We found that contradictory hallucinations are easier to detect with an AUROC of 0.79, compared to 0.69 for unsupported ones. This makes intuitive sense — when the AI contradicts the context, the probability distribution shifts more dramatically, producing a stronger KL divergence signal."*

Key numbers to remember:

Contradictory AUROC: 0.7905 | Unsupported AUROC: 0.6861 | Gap: 0.1044 | KL is best metric for both

Experiment 6 — AUROC by Generator Model

What to say:

> *"RAGTruth contains responses from 6 different language models. We computed our composite AUROC separately for each generator. Our metric works best on Mistral-7B at 0.816 and GPT-3.5 at 0.814, and least well on Llama-2-7B at 0.664. Smaller models like Llama-2-7B produce less distinctive probability distributions, which makes the uncertainty signal weaker and harder to detect."*

Key numbers to remember:

Best: mistral-7B (0.8164) | Worst: llama-2-7b (0.6644) | GPT-4: 0.7269

Experiment 7 — Failure Cases

What to say:

> *"We identified 3 false negatives — hallucinations our metric missed — and 3 false positives — clean answers our metric incorrectly flagged. The false negatives were mostly from Mistral and Llama-2-70B, which produce well-calibrated distributions with smaller KL shifts even when hallucinating. The false positives were mostly from GPT-4, which generates factually dense responses with high token-level variation, producing elevated KL scores even when faithful."*

Experiment 8 — SOTA Gap

What to say:

> *"We compared our method against published state-of-the-art systems. Our composite closes 53.2% of the gap between the entropy baseline and LUMINA, the best published supervised method. The formula is: our AUROC minus entropy baseline, divided by LUMINA minus entropy baseline, times 100. That gives us 0.7345 minus 0.5806, divided by 0.8700 minus 0.5806, which equals 53.2%. This is above the 50% threshold required for full marks."*

Key numbers to remember:

SOTA gap closed: 53.2% | LUMINA: 0.87 | ReDeEP: 0.82 | Our composite: 0.7345

4. Demo Walkthrough — Sample Input and Expected Output

Before the Evaluation Session

1. Open Command Prompt and run:

```
cd C:\Nlp_Assignment nlp_env\Scripts\activate jupyter notebook
```

2. Open 05_demo.ipynb
3. Run Cell 1 — loads GPT-2 and saved model (takes 1-2 minutes)
4. Run Cell 2 — defines the pipeline function
5. Leave Cell 3 empty and ready for evaluator input

Sample Input — Hallucinated Answer (Should Detect)

Use this example to test and demonstrate before the evaluator gives their input:

```
question = "When did Anne Frank die?" context = """Anne Frank died in the Bergen-Belsen concentration camp in February 1945, likely around February 12.""" answer = "Anne Frank died on February 7, 2022, according to records."
```

What to say when showing this:

> *"Here we give our pipeline a question with retrieved context and a hallucinated answer. The answer says 2022 but the context clearly says 1945. Watch how our metrics respond."*

Sample Input — Faithful Answer (Should Not Flag)

```
question = "What is the capital of Japan?" context = """Japan is an island country in East Asia. Its capital city is Tokyo, which is also its largest city.""" answer = "The capital of Japan is Tokyo."
```

What to say when showing this:

> *"Now we test with a faithful answer that directly matches the context. The composite score should be low and the prediction should show faithful."*

Understanding the Output

Output Field	What It Means	Good Sign
Entropy Only	Model uncertainty without context	Higher = more uncertain
Information Gain	Entropy drop when context added	Positive = context helped
KL Divergence	Distribution shift with context	Higher = bigger shift
Confidence Drop	Change in max probability	Negative = more confident with context
Composite Score	Final hallucination probability	>0.5 = hallucinated, <0.5 = faithful
Token IG Scores	Per-token information gain	Negative tokens = suspicious positions

5. Likely Questions from the Evaluator — and Answers

THE GUARANTEED QUESTION — Memorize This

The rubric explicitly states the evaluator will ask this without warning:

> *"Explain exactly how KL divergence is computed in your code, and show me where in the code it happens."*

Answer to give:

> *"KL divergence measures how different the model's token probability distribution is when context is added versus when it is not. For each token position, we run the model twice — once with context and once without. We get a probability distribution over the full vocabulary of 50,257 tokens for each run. We then compute KL divergence between these two distributions using scipy's kl_div function, sum across the vocabulary to get the total KL for that token, and average across all token positions to get a single score for the answer. The code is in notebook 02_core_pipeline inside the compute_all_metrics function."*

Q: Why did you choose GPT-2?

> *A: "GPT-2 gives us direct access to token-level probability distributions without requiring GPU infrastructure. Our approach is model-agnostic — the methodology works with any language model. We chose GPT-2 specifically because it runs on CPU and is fast enough for our experiments."*

Q: Why is your AUROC 0.73 and not higher?

> *A: "0.73 is competitive for an unsupervised method using only a small model. LUMINA achieves 0.87 but uses supervised training with labels at inference time. Our method requires no labels, which is a significant practical advantage. We close 53.2% of the gap to LUMINA without any supervision."*

Q: What is AUROC?

> *A: "AUROC stands for Area Under the ROC Curve. It measures how well our system separates hallucinated from non-hallucinated samples. A score of 1.0 is perfect — it means we correctly rank every hallucinated sample above every clean sample. A score of 0.5 is random guessing. Our composite score of 0.7345 means we correctly rank 73.45% of hallucinated-clean pairs."*

Q: Why did you use 500 samples and not all 15,000?

> *A: "Our professor confirmed that a stratified 500-sample subset is acceptable given CPU compute constraints. The subset is drawn from the held-out test split with equal numbers of hallucinated and clean samples. We report the subset size explicitly in every result table as required."*

Q: What is the difference between contradictory and unsupported hallucinations?

> *A: "Contradictory hallucinations directly conflict with the retrieved context — for example, saying a date is 2022 when the context says 1945. Unsupported hallucinations introduce information not present in the context at all — for example, adding details that were never mentioned. Contradictory ones are easier to detect because they produce stronger KL divergence signals."*

Q: What is bootstrap confidence interval?

A: "Bootstrap CI is a statistical method to estimate the reliability of our AUROC score. We randomly resample our 500 test samples with replacement 1000 times and compute AUROC each time. The 95% CI is the range containing 95% of these bootstrap AUROCs. Our composite CI of [0.689-0.779] means we are 95% confident the true AUROC lies in this range."

Q: What is SelfCheckGPT?

A: "SelfCheckGPT is a published hallucination detection method by Manakul et al. 2023. It detects hallucinations by checking consistency across multiple samples of the same generation. We implemented a lightweight version using prompt variation consistency as our second baseline. It achieves AUROC 0.6805 on our test set, which our composite beats."

Q: Why did HaluEval perform better than RAGTruth?

A: "HaluEval QA samples tend to have clearer, more direct question-answer pairs with straightforward context. This produces more distinctive uncertainty signals compared to RAGTruth which has mixed task types including summarization and data-to-text. The simpler structure of HaluEval makes the with-context versus without-context difference more pronounced."

Q: Can you explain the temporal analysis?

A: "We used the character-level span annotations in RAGTruth to find the exact token position where each hallucination starts. We call this position t. We then measured all our metrics at positions t-3, t-2, t-1, t, and t+1. The goal is to see if our metrics rise before the hallucination occurs. We found KL divergence shows an elevated signal at t-3 which is 3 tokens before the hallucination — suggesting pre-generation uncertainty elevation."

Q: What would you do differently with more compute?

A: "We would use a larger instruction-tuned model like Llama-2-7B as our base instead of GPT-2. Larger models produce more meaningful probability distributions which would likely improve our AUROC significantly. We would also implement true semantic entropy using sentence clustering, and run on the full 15,000 sample dataset for more stable results."

6. Team Roles During Evaluation

The evaluator may ask ANY team member to explain ANY part. Everyone must understand the basics. Here is the suggested split for who leads which part:

Person	Lead Responsibility	Must Know
Zayaan	Live demo + KL divergence code explanation	All 5 metrics, composite model, how to run 05_demo.ipynb
Ridhwan	Slides presentation + dataset explanation	RAGTruth vs HaluEval, what hallucination types are, Exp 4 results
Aaqib	Experiment results explanation	AUROC meaning, temporal analysis, SOTA gap calculation
Mariah	References and theoretical justification	What SelfCheckGPT is, what KL divergence is, why uncertainty = hallucination signal

Critical Reminders

- Open Jupyter and run Cell 1 and Cell 2 of 05_demo.ipynb BEFORE your slot starts.
- The evaluator will give you an unseen input — just type it into Cell 3 and press Shift+Enter.
- If asked to show KL divergence code — open 02_core_pipeline.ipynb and scroll to compute_all_metrics.
- All result numbers are in the submitted report. Do not guess numbers — refer to the report.
- If you do not know an answer, say: "Let me refer to our implementation" and open the relevant notebook.
- Do not panic if the demo shows FAITHFUL for a hallucinated input — explain it as a known limitation of GPT-2.

Good luck to the whole team! You have done solid work — present it confidently.